Stable rank based expressivity and generalization analysis of Low-Rank-Adapters

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MIPT

2025

Introduction

LoRA – most popular yet still understudied method of parameter-efficient fine-tuning

Problem

Existing theoretical estimates rely on algebraic rank and misalign with practical results

Observation

Empirical results show that it is not the rank that limits the performance of the adapters or lead to overfitting

Proposal

Derive the more practical bounds based on stable rank

Preliminary theory

Interpretable SR-Constrained Approximation Error Lower Bound

For matrix A stable rank is $sr(A) = \dfrac{\|A\|_F^2}{\|A\|_{op}^2} = \dfrac{\sum_i \sigma_i^2(A)}{\sigma_1^2(A)}$

Lemma (single matrix approximation lower bound)

A – target matrix, $\hat{A}$ - an estimate. $S_A = \mathrm{sr}(A),\ S_{\hat{A}} = \mathrm{sr}(\hat{A}),\ S_{\hat{A}} \leqslant S_A$ then:

$$\|A - \hat{A}\|_F \geqslant \sigma_1(A) \left(\frac{\sqrt{S_A} - \sqrt{S_{\hat{A}}}}{1 + \sqrt{S_{\hat{A}}}} \right)$$

Lemma 2 (Optimal SR-constrained approximation)

Let $A_R = \sum_{i=1}^{R} \sigma_i u_i v_i^T$ be the optimal rank-R approx. of A, $S \in [1, R]$ the SR limit. $\mathcal{M}_{R,S}$ = matrices with rank $\leqslant R$, sr $\leqslant S$.
If $\mathrm{sr}(A_R) \leqslant S$: $\hat{A} = A_R$. If $\mathrm{sr}(A_R) > S$, the optimal $\hat{A} \in \mathcal{M}_{R,S}$ has:

$$\hat{\sigma}_1 = \frac{\boldsymbol{\sigma_1} + \sqrt{S-1}\sqrt{\sum_{j=\mathbf{2}}^{R} \sigma_j^{\mathbf{2}}}}{S}, \quad \hat{\sigma}_i = \gamma \sigma_i \ (i \geqslant 2),\ \gamma = \frac{\sqrt{S-1}\hat{\sigma}_{\mathbf{1}}}{\sqrt{\sum_{j=\mathbf{2}}^{R} \sigma_j^{\mathbf{2}}}}$$

Minimal error: $\|A - \hat{A}\|_F^2 = \sum_{i=R+1}^{D} \sigma_i^2 + \Delta_S$,

$\Delta_S = \frac{1}{S}\left(\sqrt{\sum_{i=2}^{R} \sigma_i^2} - \sqrt{S-1}\sigma_1\right)^2$ (If $\mathrm{sr}(A_R) \leqslant S$, $\Delta_S = 0$)

Fully connected network approximation

Notation

target FNN $\quad \overline{f} := FNN_{\overline{L},D}(.;(\overline{W}_l)_{l=1}^{L},(\overline{b}_l)_{l=1}^{L})$,

frozen FNN $\quad f_0 := FNN_{L,D}(.;(W_l)_{l=1}^{L},(b_l)_{l=1}^{L})$,

adapted FNN $\quad f := FNN_{L,D}(.;(W_l+\Delta W_l)_{l=1}^{L},(\hat{b}_l)_{l=1}^{L})$

$b_l, \overline{b}_l, \hat{b}_l \in \mathbb{R}^D$, $W_l, \overline{W}_l, \Delta W_l \in \mathbb{R}^{D\times D}$, but $\Delta W_l = A_l B_l$, where $A_l \in \mathbb{R}^{D\times R}$ and $B_l \in \mathbb{R}^{R\times D}$ and $R \leqslant D$ – the LoRA rank. Additionally, we assume and $\overline{L} = L$.

Theorem (FNN error bound)

Let $x \in \mathbb{R}^d$ be the random input with covariance matrix $\Sigma = \mathbb{E}[xx^T]$ Let $E_l = \overline{W}_l - W_l$ be the weight discrepancy. Let $E_{l,R}$ be its optimal rank-R approx. with singular values $\sigma_{l,1} \geqslant \cdots \geqslant \sigma_{l,R}$. Assume $\hat{b}_l = \overline{b}_l$. Then there exist adapters ΔW_l with $\mathrm{sr}(\Delta W_l) \leqslant S \quad \forall l \in \overline{1,L}$ such that the expected error is bounded by:

$$\mathbb{E}\|f(\mathrm{x}) - \overline{f}(\mathrm{x})\|_2 \leqslant \beta \sum_{l=1}^{L} \left(\prod_{k=l+1}^{L} \|\overline{W}_k\|_2 \right) \mathcal{T}_l, \text{ where}$$

$\beta = \max_{i\in[L]} \left(\left(\prod_{j=1}^{i} \|\overline{W}_j\| \right) \sqrt{\mathrm{tr}(\Sigma)} + \sum_{j=1}^{i} \left(\prod_{k=j+1}^{i} \|\overline{W}_k\| \right) \|\overline{b}_j\| \right)$

and $\mathcal{T}_l = \sqrt{\sum_{j=R+1}^{D} \sigma_{l,j}^2 + \Delta_{S,l}}, \quad$ where

$\Delta_{S,l} = \frac{1}{S}\left(\sqrt{\sum_{j=2}^{R} \sigma_{l,j}^2} - \sqrt{S-1}\sigma_{l,1} \right)^2$ if $\mathrm{sr}(E_{l,R}) > S, \quad 0$ else

Transformer Network Approximation

Notation

$TN_{L,D}$: L blocks, H-head self-attention + 2-layer FFN, no dropout/norm.
Target $\overline{f}$, frozen f_0, adapted f (LoRA on all Q, K, V, O, both FFN layers, output).
$\mathcal{T}(G) = \sqrt{\sum_{i=R+1}^{D} \sigma_i^2(G) + \Delta_S(G)}$ – SR error from **Lemma 2**.

Theorem (Transformer error bound)

There exists a $\leqslant S$ SR adapter configuration such that

$$\mathbb{E}\|f(x) - \overline{f}(x)\|_F \leqslant \beta \sum_{l=1}^{L} \Big(\prod_{k=l+1}^{L} \gamma_k \Big) \mathcal{D}_l$$

- γ_k: Lipschitz constant of target block k w.r.t. input (via FFN + MHA)
- $\mathcal{D}_l = \mathcal{T}_{2,l} + \|\overline{W_{2,l}}\|\mathcal{T}_{1,l} + \|\overline{W_{2,l}}\|\|\overline{W_{1,l}}\| \sum_h \mathcal{D}^h_{att,l}$ (FFN + attention errors)
- $\mathcal{D}^h_{att,l}$: per-head distortion $\propto \mathcal{T}_O, \mathcal{T}_V, \mathcal{T}_Q, \mathcal{T}_K$ amplified by score sensitivity C_{score}

Upper-bound for LoRA Generalization Gap

Notation

Depth-T network, 1-Lipschitz $\sigma(.)$, $\sigma(0) = 0$, LoRA on each layer $\Delta W_t = B_t A_t^T$.
$\|x_i\|_2 \leqslant B_x$, $\|\Delta W_t\|_{op} \leqslant \Lambda_t$, $\mathrm{sr}(\Delta W_t) \leqslant S_t$, $d_{max} = \max_t(d_t)$.

$$\text{Margin: } m(f_W(x), y) = f_W(x)_y - \max_{j \neq y} f_W(x)_j$$

$$L_\gamma(W) = \mathbb{P}_{(x,y)}[m(f_W(x), y) \leq \gamma], \quad \widehat{L}_\gamma(W) = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}[m(f_W(x_i), y_i) \leq \gamma]$$

Theorem (PAC-Bayesian generalization bound)

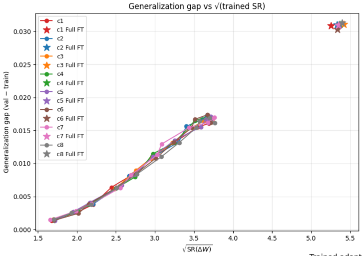
$\forall \gamma > 0$ and any $W = W_0 + \Delta W$ that satisfies the aforementioned conditions with probability at least $1 - \delta$ by the choice of training sample of size N:

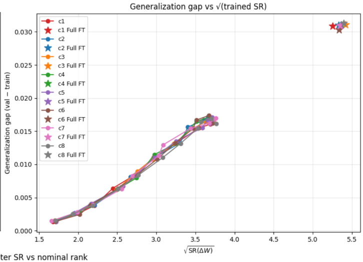
$$L_0(W) \leqslant \widehat{L}_\gamma(W) + \mathcal{O}\left(\sqrt{\frac{B_x^2 T^2 (d_{max} + \ln(TN)) \left(\prod_{t=1}^{T} \bar{\beta}_t^2\right) \sum_{t=1}^{T} \frac{\Lambda_t^2 S_t}{\bar{\beta}_t^2} + \log(N/\delta)}{\gamma^2 N}}\right)$$

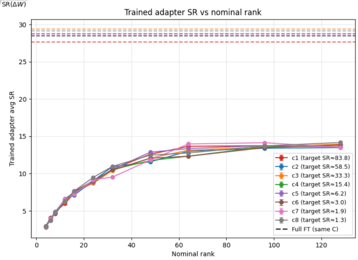
Scales as $\mathcal{O}(\sqrt{S/N})$ unlike previously known best $\mathcal{O}(\sqrt{r/N})$.

Experiments
Controlled small scale setting

Initialize 10-layer MLP $\overline{f}_0$ (teacher), copy it's weights to f_0 (student)
Shift $\overline{f}_0 \to \overline{f}$ by adding $\Delta\overline{W}_i \in \mathbb{R}^{d\times d}$ with variable SR to the layers
Generate data $x \sim \mathcal{N}(0, I) \to \overline{f} \to y$. Train $f_0 \to f$ on (x, y) pairs and evaluate.







Conclusion

1. Optimal rank $\leqslant R$ stable rank $\leqslant S$ lemma proven
2. FNN and Transformer previously known expressivity bounds adjusted for constrained stable rank
3. Generalisation gap upper bound tightened from $\mathcal{O}(\sqrt{R/N})$ to $\mathcal{O}(\sqrt{S/N})$
4. Stable rank empirically shown to be a better proxy for generalization and expressivity scaling